Fundamentals Physics

Eleventh Edition

Halliday

Chapter 16

Waves - I

16-1 Transverse Waves

Learning Objectives

16.01 Identify the three main types of waves.

16.02 Distinguish between transverse waves and longitudinal waves.

16.03 Given a displacement function for a traverse wave, determine amplitude y_m, angular wave number k, angular frequency ω, phase constant ϕ, and direction of travel, and calculate the phase $kx \pm t + \phi$ and the displacement at any given time and position.

16-1 Transverse Waves (2 of 11)

16.04 Given a displacement function for a traverse wave, calculate the time between two given displacements.

16.05 Sketch a graph of a transverse wave as a function of position, identifying amplitude y_m wavelength.

λ, where the slope is greatest, where it is zero, and

where the string elements have positive velocity, negative velocity, and zero velocity.

16.06 Given a graph of displacement versus time for a transverse wave, determine amplitude y_m and period T.

16-1 Transverse Waves (3 of 11)

16.07 Describe the effect on a transverse wave of changing phase constant ϕ.

16.08 Apply the relation between the wave speed v, the distance traveled by the wave, and the time required for that travel.

16.09 Apply the relationships between wave speed v, angular frequency ω, angular wave number k, wavelength λ, period T, and frequency f.

16-1 Transverse Waves (4 of 11)

16.10 Describe the motion of a string element as a transverse wave moves through its location, and identify when its transverse speed is zero and when it is maximum.

16.11 Calculate the transverse velocity $u(t)$ of a string element as a transverse wave moves through its location.

16.12 Calculate the transverse acceleration $a(t)$ of a string element as a transverse wave moves through its location.

16.13 Given a graph of displacement, transverse velocity, or transverse acceleration, determine the phase constant ϕ.

16-1 Transverse Waves (5 of 11)

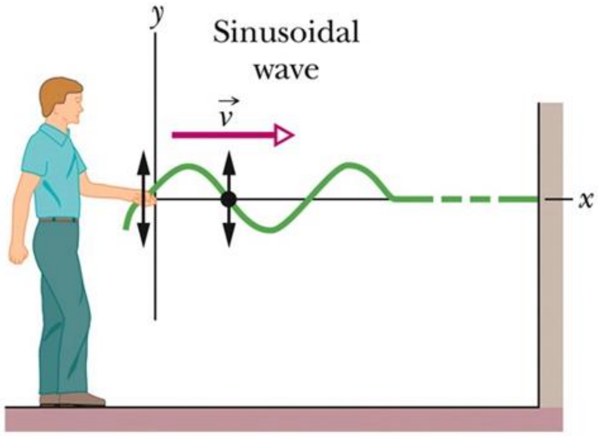
Types of Waves

1. **Mechanical Waves:** They are governed by Newton's laws, and they can exist only within a material medium, such as water, air, and rock. Examples: water waves, sound waves, and seismic waves.
2. **Electromagnetic waves:** These waves require no material medium to exist. Light waves from stars, for example, travel through the vacuum of space to reach us. All electromagnetic waves travel through a vacuum at the same speed c = 299 792 458 m/s.
3. **Matter waves:** These waves are associated with electrons, protons, and other fundamental particles, and even atoms and molecules. Because we commonly think of these particles as constituting matter, such waves are called matter waves.

16-1 Transverse Waves (6 of 11)

Transverse and Longitudinal Waves

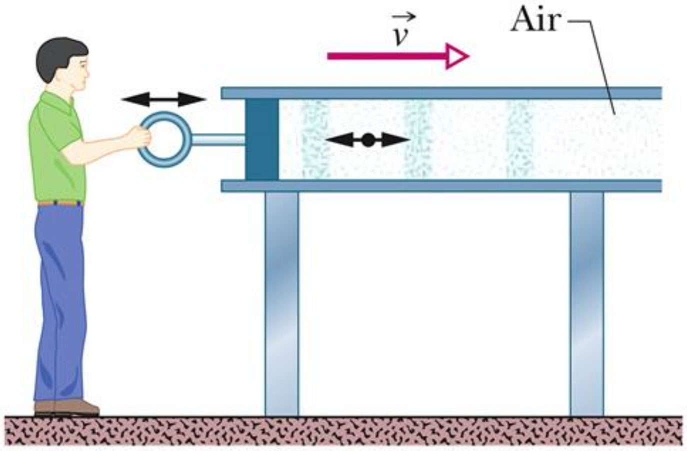
A sinusoidal wave is sent along the string (Figure (a)). A typical string element moves up and down continuously as the wave passes. This is **transverse wave.**



(a) Transverse Wave

16-1 Transverse Waves (7 of 11)

A sound wave is set up in an air- filled pipe by moving a piston back and forth (Figure (b)). Because the oscillations of an element of the air (represented by the dot) are parallel to the direction in which the wave travels, the wave is a **longitudinal wave.**

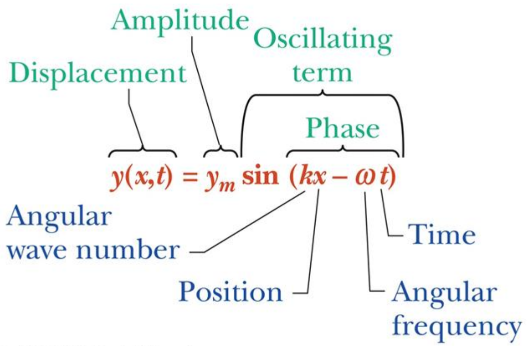


(b) Longitudinal Wave

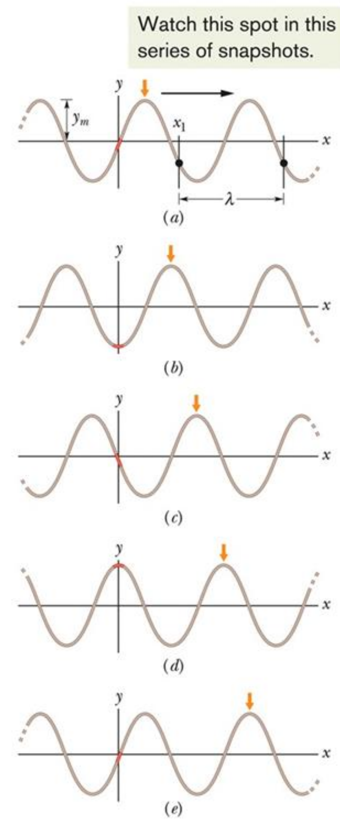
16-1 Transverse Waves (8 of 11)

Sinusoidal Function

Five “snapshots” (y vs x each at a constant time) of a string wave traveling in the positive direction along an x axis. The amplitude y_m is indicated. A typical wavelength λ, measured from an arbitrary position x_1, is also indicated.



The sine function describes the shape of the wave



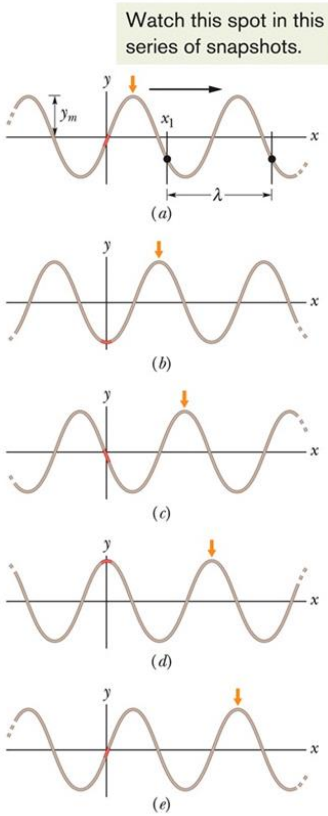
16-1 Transverse Waves (9 of 11)

Period, Wave Number, Angular Frequency and Frequency

$$k = \frac{2\pi}{\lambda} \quad \text{(angular wave number).}$$

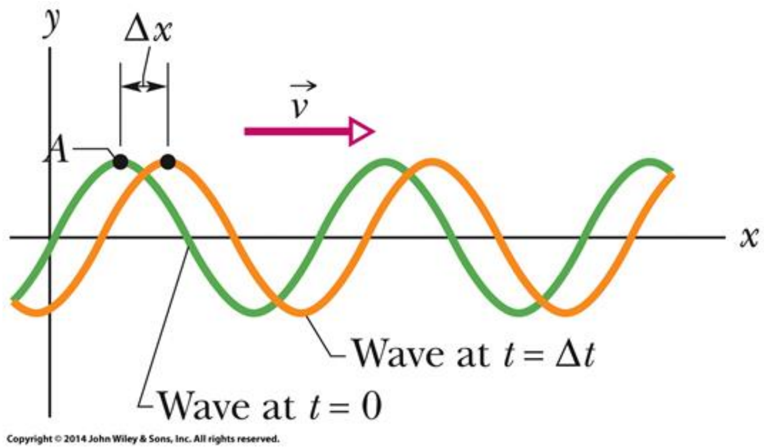
$$\omega = \frac{2\pi}{T} \quad \text{(angular frequency).}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi} \quad \text{(frequency).}$$



16-1 Transverse Waves (10 of 11)

The Speed of a Traveling Wave



Two snapshots of the wave: at time $t = 0$, and then at time $t = \Delta t$. As the wave moves to the right at velocity v, the entire curve shifts a distance Δx during Δt.

16-1 Transverse Waves (11 of 11)

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f \qquad (\text{wave speed}).$$

$$y(x,t) = y_m \sin(kx + \omega t).$$

16-5 Interference of Waves (1 of 6)

Learning Objective

16.18 Apply the principle of superposition to show that two overlapping waves add algebraically to give a resultant (or net) wave.

16.19 For two transverse waves with the same amplitude and wavelength and that travel together, find the displacement equation for the resultant wave and calculate the amplitude in terms of the individual wave amplitude and the phase difference.

16-5 Interference of Waves (2 of 6)

16.20 Describe how the phase difference between two transverse waves (with the same amplitude and wavelength) can result in fully constructive interference, fully destructive interference, and intermediate interference.

16.21 With the phase difference between two interfering waves expressed in terms of wavelengths, quickly determine the type of interference the waves have.

16-5 Interference of Waves (3 of 6)

Principle of Superposition of waves

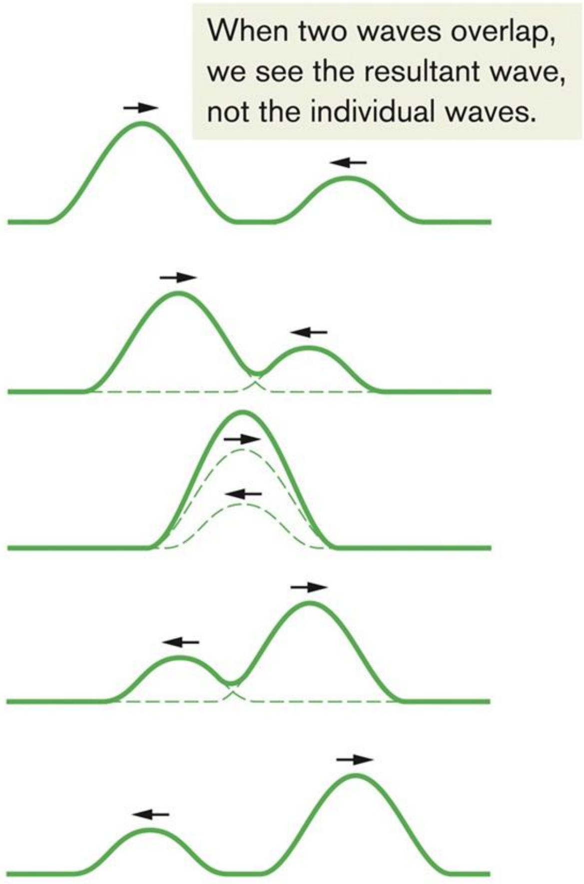
Let $y_1(x, t)$ and $y_2(x, t)$ be the displacements that the string would experience if each wave traveled alone. The displacement of the string when the waves overlap is then the algebraic sum

$$y'(x,t) = y_1(x,t) + y_2(x,t).$$

This summation of displacements along the string means that

Overlapping waves algebraically add to produce a **resultant wave** (or **net wave**).

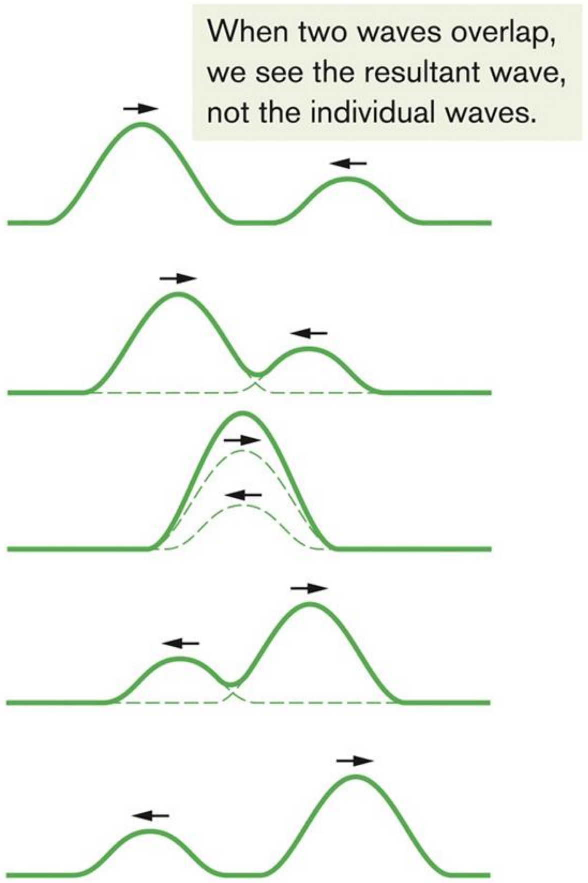
Overlapping waves do not in any way alter the travel of each other.



16-5 Interference of Waves (4 of 6)

The resultant wave due to the interference of two sinusoidal transverse waves, is also a sinusoidal transverse wave, with an amplitude and an oscillating term.

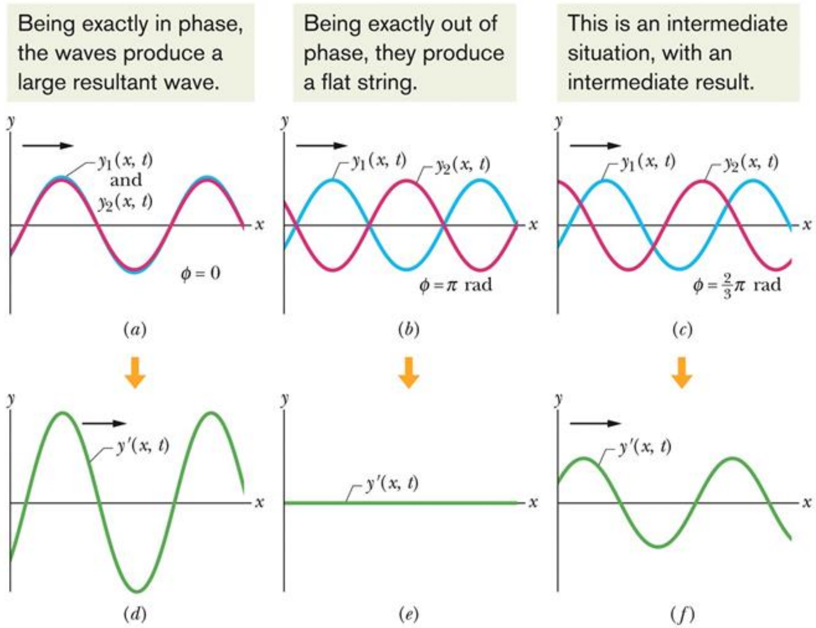
$$\overbrace{y'(x,t)}^{\text{Displacement}} = \underbrace{[2y_m \cos \tfrac{1}{2}\phi]}_{\substack{\text{Magnitude}\\ \text{gives}\\ \text{amplitude}}} \underbrace{\sin(kx - \omega t + \tfrac{1}{2}\phi)}_{\substack{\text{Oscillating}\\ \text{term}}}$$



16-5 Interference of Waves (5 of 6)

Constructive and Destructive Interference

$$y'(x,t) = y_1(x,t) + y_2(x,t).$$



16-5 Interference of Waves (6 of 6)

Two identical sinusoidal waves, $y_1(x, t)$ and $y_2(x, t)$, travel along a string in the positive direction of an x axis. They interfere to give a resultant wave $y'(x, t)$. The resultant wave is what is actually seen on the string. The phase difference ϕ between the two interfering waves is (a) 0 rad or 0°, (b) π rad or 180°, and (c) $\frac{2}{3}\pi$ rad or 120°. The corresponding resultant waves are shown in (d), (e), and (f).

Table 16-1 Phase Difference and Resulting Interference Types[a]

Phase Difference, in			Amplitude of Resultant Wave	Type of Interference
Degrees	Radians	Wavelengths		
0	0	0	$2y_m$	Fully constructive
120	$\frac{2}{3}\pi$	0.33	y_m	Intermediate
180	π	0.50	0	Fully destructive
240	$\frac{4}{3}\pi$	0.67	y_m	Intermediate
360	2π	1.00	$2y_m$	Fully constructive
865	15.1	2.40	$0.60y_m$	Intermediate

[a]The phase difference is between two otherwise identical waves, with amplitude y_m, moving in the same direction.

Summary (1 of 4)

Waves

- Transverse Waves
- Longitudinal Waves

Wave Speed

- $\dfrac{\text{Angular velocity}}{\text{Angular wave number}}$

$$v = \frac{\omega}{k} = \frac{\lambda}{T} = \lambda f. \qquad \textbf{Equation (16-13)}$$

Summary (2 of 4)

Sinusoidal Waves

- Wave moving in positive direction (vector)

$$y(x,t) = y_m \sin(kx - \omega t)$$ **Equation (16-2)**

Traveling Waves

- A functional form for traveling waves

$$y(x,t) = h(kx \pm \omega t)$$ **Equation (16-17)**

Summary (3 of 4)

Powers

- Average Power is given by

$$P_{\text{avg}} = \tfrac{1}{2}\mu v\omega^2 y_m^2$$ **Equation (16-33)**

Standing Waves

- The interference of two identical sinusoidal waves moving in opposite directions produces standing waves.

$$y'(x,t) = [2y_m \sin kx]\cos\omega t.$$ **Equation (16-60)**

Summary (4 of 4)

Interference of Waves

- Two sinusoidal waves on the same string exhibit interference

$$y'(x, t) = [2y_m \cos \tfrac{1}{2}\phi] \sin(kx - \omega t + \tfrac{1}{2}\phi).$$ **Equation (16-51)**

Resonance

- For a stretched string of length L with fixed ends, the resonant frequencies are

$$f = \frac{v}{\lambda} = n\frac{v}{2L}, \qquad \text{for } n = 1, 2, 3, \ldots$$ **Equation (16-66)**